Harsh Event Detection

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The Samsara Vehicle Gateway (VG) detects harsh events, such as harsh brakes, accelerations, turns, and crashes. After installation, Samsara assigns harsh event thresholds based on the detected vehicle type. On supported gateways and CM3x dash cams, harsh event detection applies your sensitivity choices on the device using calibrated, vehicle-oriented sensor data. Vehicles with CM11, CM12, or CM22 cameras, some gateways, or fleets using a Diagnostic Hub (ACC-BDH or ACC-BDE) use the legacy harsh event detection and sensitivity experience. Each harsh event surfaces in Harsh Event Reports and for Reviewing Safety Events in the Safety Inbox.

Crash detection identifies vehicle collisions using Vehicle Gateway sensor data, a backend machine-learned model, and—in some cases—forward-facing camera video from harsh brake events. Confirmed crashes surface in the **Safety Inbox** during and after trips when the Vehicle Gateway remains powered on. Crash events pass through **Advanced Crash Filtering** before alerts, coaching workflows, and inbox visibility unless your organization has opted out.

To raise a harsh event in the Samsara dashboard, the following minimum criteria must be met:

- **Trip requirements depend on the event type.**
  Harsh braking, harsh acceleration, and harsh turning events surface only when a vehicle is on an active trip. A trip begins when GPS speed exceeds 5 mph and ends after GPS speed remains below 5 mph for five minutes.
  Crashes are evaluated differently from other safety events:
  - Crashes can be detected both during and outside of an active trip, as long as the Vehicle Gateway is powered on and the camera is recording.
  - Crashes that occur after a trip has ended—while the Vehicle Gateway remains powered on (for example, when a parked vehicle is struck or during a low-speed yard maneuver)—are also detected and surfaced in the **Safety Inbox**.
  - Crashes that occur after the Vehicle Gateway has fully powered down are not captured for **Safety Inbox** reporting.

> **Note**
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> Off-trip crashes are not detected when your fleet uses engine-state recording. Off-trip crashes cannot be coached because driver assignment requires an active trip.

- **The VG detects an event with a g-force above the configured threshold.**
  When the device is calibrated, harsh event detection on supported CM3x and gateway combinations uses oriented acceleration for harsh acceleration, harsh brake, and harsh turn detection (with smoothing to reduce noise). Crash detection uses the same orientation but does not apply that smoothing so high-frequency collision signatures are retained. If the gateway or camera combination uses legacy detection, harsh event detection still uses oriented data after calibration. For details about harsh event types and thresholds, see Harsh Event Types and Thresholds.
- **The safety event has met the minimum safety event criteria.**
  For safety event criteria, see Reviewing Safety Events in the Safety Inbox.
- **The safety event video is available to review.**
  For video retrieval limitations and considerations, see Video Retrieval Process.
  If no dash cam footage is available for an event, the harsh event does not appear in the **Safety Inbox**; related information may still appear in incident reporting and driver safety scoring.

Get Started

To get started with Harsh Event Detection, review the following workflows to optimize its use and customize it for your fleet:

- **Review Event Detection Best Practices**: For the most accurate Harsh Event Detection, review the Harsh Event Best Practices.
- **Customize Event Detection**: Samsara automatically applies recommended settings based on the detected vehicle type. To customize the vehicle type and configure detection sensitivity for harsh acceleration, braking, and turning thresholds, see Configure Harsh Event Settings.
- **Set alerts**: In addition to in-cab audio alerts for drivers, you can also configure an alert to receive an email notification or an SMS when a harsh event occurs.
- **Enable High Speed Power Loss detection**: To surface severe incidents when a Vehicle Gateway loses power at high speed, see Enable Hi Power Loss Detection in Configure Harsh Event Settings.

For the most accurate harsh event detection that's appropriate for your fleet, Samsara recommends that you:

- Ensure the VG is securely installed in a location where the device is unlikely to be disturbed by the driver. For example, if the VG is loosely installed or kicked by the driver this could interfere with detection or generate a false harsh event.
- Only relocate the device if necessary. After you reposition the gateway, it re-calibrates. Small orientation changes can be absorbed while calibration stays valid; a large change in how gravity reads on the sensor forces a reset until the vehicle drives enough varied motion to re-lock orientation.
- Select an appropriate Trip Activity threshold for your fleet.
- Verify device calibration status is `Calibrated` from the Gateways report. For proper calibration, ensure the vehicle has adequate drive time with varied driving styles including braking, accelerating, and turning. If your **Device Calibration Status** is not `Calibrated`, see Calibration Status Types.

Calibration Status Types

The calibration process can take from a few minutes up to a few hours and depends on gateway health, driving behavior and duration, and successful installation. The gateway calibrates so sensor readings align with how the vehicle moves. When the gateway supports on-device harsh event detection and calibration completes, harsh acceleration, braking, and turn detection use that pipeline. If the device is not eligible for on-device detection, legacy harsh event detection runs until calibration and eligibility requirements are met. Crash detection begins once the device has enough calibration data to identify impacts.

The gateway device list (**Settings** (⚙) > **Devices** > **Gateways**) indicates one of the following **Device Calibration Status** for your VG:

Device Calibration Status	Description	Recommended Action
N/A	Device does not require calibration for detection.	—
Pending	Device is updating to enable calibration.	Ensure that the device is powered on and allow up to two hours for the update process to complete.
Not Calibrated	Calibration process has not yet started.	Ensure the vehicle is active and review the Best Practices to ensure successful calibration.
Calibrating	Calibration is in process.	Review the Best Practices to ensure successful calibration.
Calibrated	Calibration is complete.	—

Configure Harsh Event Settings

The VG automatically detects the vehicle type and assigns harsh event thresholds. You can influence the volume of events that the VG surfaces to the Samsara dashboard by modifying the default sensitivity settings. These settings can be modified for all vehicles in your fleet or for selected vehicles.

To configure settings for individual vehicles, see Configure Safety Settings for Individual Vehicles or use the following workflow to configure settings for all vehicles in your fleet:

1. Alternatively, select the Settings icon (⚙) at the bottom of your Fleet menu.
2. From within the Fleet settings, select **Safety > Harsh Events**.
3. Select the default **Vehicle Type** for vehicles in your fleet (**Automatic**, **Passenger Car**, **Light Truck**, **Heavy Duty**).
4. Adjust the **Harsh Event Sensitivity** settings for each harsh event category (**Harsh Acceleration**, **Harsh Brake**, **Harsh Turn**) according to your dash cam model:
    - Harsh Event Sensitivity Settings (CM3x)
    - Harsh Event Sensitivity Settings (CM11/CM12/CM22)
5. **Save** to confirm the settings.
6. If desired, enable Advanced Crash Filtering settings.

Use Advanced Crash Filtering to delay crash visibility, alerts, and automations until crash events are reviewed in the **Safety Inbox**. This setting helps reduce false crash notifications caused by high g-force events that are not collisions. When **Require review** is enabled, crash safety events and related workflows can stay hidden until backend review completes, up to about 10 minutes. For organizations created after February 2026, this behavior is on by default unless you change it.

When **Require review** is enabled, Advanced Crash Filtering holds crash events until backend review completes, up to about 10 minutes, before they display in the **Safety Inbox** or trigger alerts and automations. This filtering reduces false crash notifications from high g-force events that are not collisions. If your organization has opted out of Advanced Crash Filtering, you may see a higher proportion of false positives than opted-in organizations. Re-enabling **Require review** is recommended.

To enable Advanced Crash Filtering for your organization:

1. Alternatively, select the Settings icon (⚙) at the bottom of your Fleet menu.
2. Select **Safety > Event Detection**.
3. Locate **Advanced Crash Filtering** and enable **Require review**.
4. **Save** to confirm the settings.

Enable High Speed Power Loss Detection

High Speed Power Loss detection surfaces incidents when a Vehicle Gateway loses power while the vehicle is traveling above a configurable speed threshold. These events display in the **Safety Inbox** with a **High Speed Power Loss** label. They provide a fallback when an impact is severe enough that the gateway cannot log a standard crash event. High Speed Power Loss is separate from threshold-based crash detection, works on all Vehicle Gateway models, and does not require a dash cam. The detection is off by default.

To enable High Speed Power Loss detection for your organization:

1. Alternatively, select the Settings icon (⚙) at the bottom of your Fleet menu.
2. From within the Fleet settings, select **Safety > Harsh Events**.
3. Enable **High Speed Power Loss** and configure the **Speed Threshold** and **Duration** values for your fleet.
4. **Save** to confirm the settings.

To receive notifications when a High Speed Power Loss event occurs, configure an alert with the **A safety event occurred** trigger and select **High Speed Power Loss** as the behavior. Remove **All Drivers** from the driver filter so the alert fires even when no driver is assigned.

Note

High Speed Power Loss events may include limited or no video because the Vehicle Gateway lost power during the incident.

Harsh Event Sensitivity Settings (CM3x)

On CM3x cameras paired with a supported gateway, harsh event sensitivity uses a 0 (off) to 100 scale in steps of 10, with optional **Advanced Options** so you can enter specific g-force values between steps or outside the slider range. Samsara recommends that you start from the recommended defaults, monitor inbox volume for one to two active weeks, then adjust. Higher values on the scale trigger at lower g-forces (more events); lower values trigger at higher g-forces (fewer events).



After you select a **Vehicle Type**, adjust **Harsh Event Sensitivity** for **Harsh Acceleration**, **Harsh Brake**, and **Harsh Turn** separately. Each slider position maps to a g-force threshold range.

When you enable **Advanced Options**, you can enter a specific g-force value between slider steps or outside the slider range. The table below lists the minimum and maximum g-force values available in Advanced Options for each event and vehicle type.

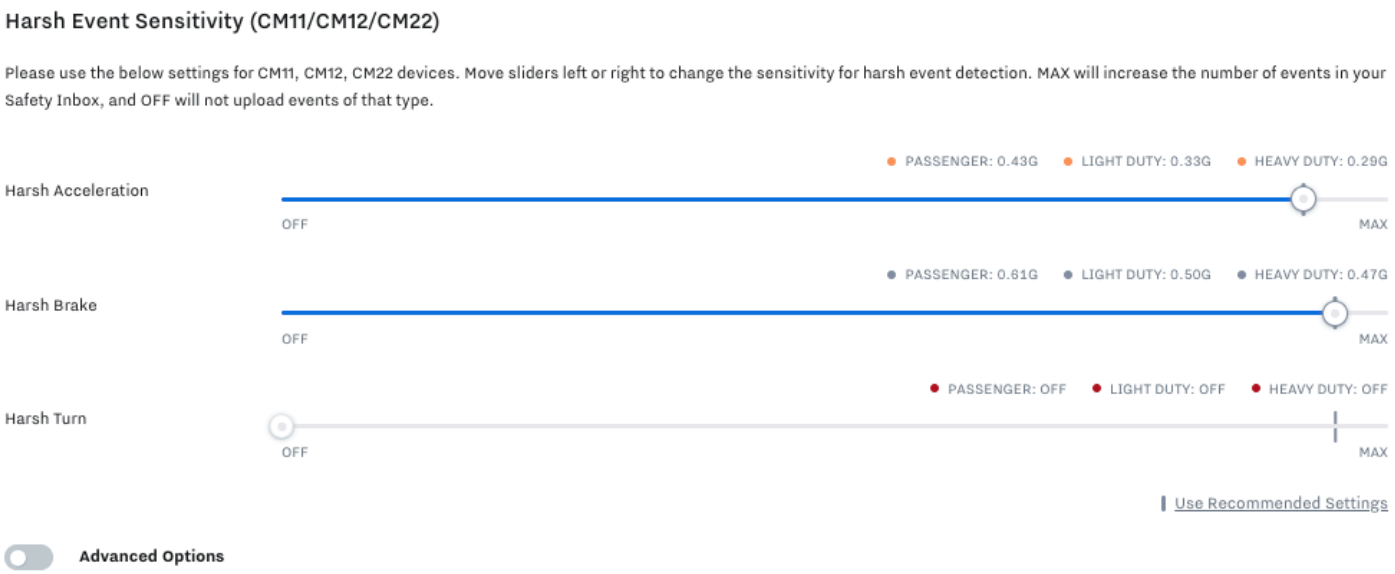
Event Type	Vehicle Type	Minimum G-Force	Maximum G-Force
Harsh Acceleration	Passenger Car	0.32 g	0.7 g
Harsh Acceleration	Light Truck	0.25 g	0.633 g
Harsh Acceleration	Heavy Duty	0.22 g	1.0 g
Harsh Brake	Passenger Car	0.3 g	1.07 g
Harsh Brake	Light Truck	0.38 g	0.995 g
Harsh Brake	Heavy Duty	0.35 g	0.899 g

Harsh Turn	Passenger Car	0.35 g	0.888 g
Harsh Turn	Light Truck	0.3 g	0.687 g
Harsh Turn	Heavy Duty	0.24 g	0.56 g

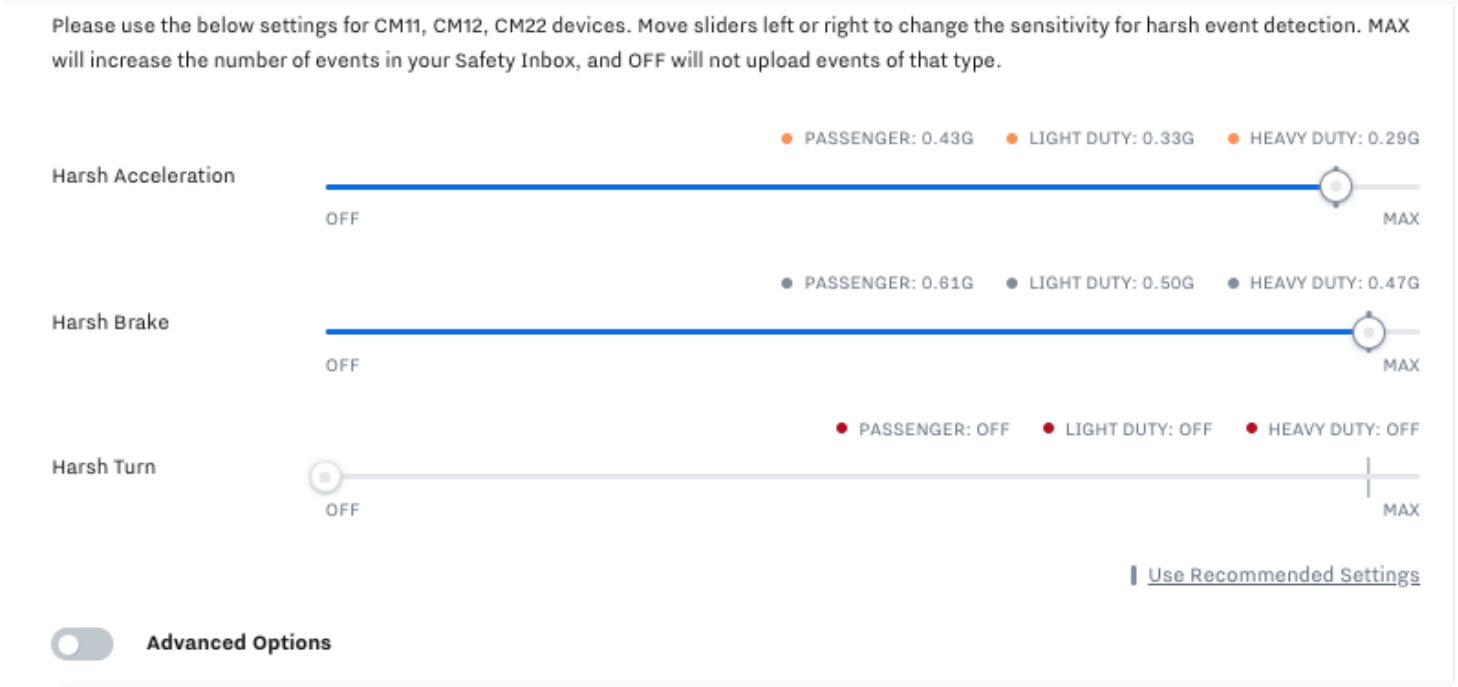
Harsh Event Sensitivity Settings (CM11/CM12/CM22)

CM11, CM12, and CM22 devices use the legacy harsh event detection and sensitivity layout, not the on-device CM3x pipeline. Harsh acceleration, brake, and turn events that occur at speeds of less than 5 mph are automatically filtered out of the **Safety Inbox**. On supported CM3x and gateway combinations, low-speed harsh events and low-speed crashes are not removed from the inbox solely because of speed.

Samsara recommends that you **Use Recommended Settings** and monitor the event volume for 1-2 active weeks before adjusting the thresholds.



When you enable **Advanced Options**, you can configure the sensitivity settings for each vehicle type. You can adjust the sensitivity from **OFF** (no events) to **MAX** for each harsh event category and vehicle type.



Harsh Event Types and Thresholds

Harsh event detection settings are configured per vehicle type and event type. On supported gateways with CM3x dash cams, you set sensitivity on a 0–100 scale (with optional advanced g-force entry) for harsh acceleration, braking, and turning. On CM11, CM12, and CM22 devices, you choose among **High**, **Normal**, and **Low** (or advanced ranges) for each behavior. The VG's built-in accelerometer detects g-forces to gauge severity relative to the vehicle class—for example, harsh brake and harsh turn thresholds differ for a passenger vehicle versus a heavy duty vehicle.

The more sensitive the setting you choose when you Configure Harsh Event Settings, the lower the g-force needed to trigger an event. Exact g-force values vary by vehicle type and by whether the vehicle uses on-device CM3x thresholds or the legacy sensitivity model. The following list summarizes the harsh event types:

Detection Logic

The VG uses calibrated, vehicle-oriented sensor data to detect harsh events. On supported CM3x and gateway combinations, harsh event detection applies additional filtering for harsh acceleration when the vehicle is stationary or moving very slowly to reduce common false positives while keeping true events. On CM11, CM12, and CM22 cameras, harsh acceleration at very low speeds may still be filtered and not generated.

Event Prioritization

When multiple events occur at the same time, the VG determines which event to surface based on event type and severity. The following logic applies:

- If a crash and one or more harsh events are detected at the same time, only the crash event is surfaced in the dashboard.
- If multiple harsh events occur simultaneously, the event with the highest g-force is surfaced.

Sensitivity Levels

Sensitivity controls how easily events are triggered based on g-force thresholds. On supported gateways with CM3x dash cams, the 0–100 scale and optional advanced g-force entry replace the High, Normal, and Low labels described below for CM11, CM12, and CM22. For slider and Advanced Options g-force ranges on CM3x, see Configure Harsh Event Settings. For those older camera models, the table still summarizes High, Normal, and Low behavior:

High	Triggers at lower g-forces (more events)
Normal	Balanced detection
Low	Triggers at higher g-forces (fewer events)

Harsh Event Types

Each harsh event type is associated with specific trigger conditions and video behavior. The list below summarizes how each event is detected and handled:

- **Crashes**: On the Vehicle Gateway, crash detection triggers when oriented acceleration magnitude exceeds about 2 g (excluding gravity), regardless of vehicle type. Each crash event surfaces a 30-second video to the **Safety Inbox**. After upload, Samsara classifies potential crashes using heuristics and a machine-learned backend model. Confirmed crashes display in the **Safety Inbox** after they pass **Advanced Crash Filtering**, unless your organization has opted out.
  Samsara also evaluates harsh brake, harsh acceleration, and harsh turn events that stay below the crash g threshold: forward-facing camera video is scanned with machine learning for collision signatures. Likely collisions are sent to Safety Event Review for human verification before customers see them. Confirmed collisions are promoted to full crash events in the **Safety Inbox** with crash alerts, scoring, and reporting, and they carry the **Promoted to Crash** context label so you can tell them apart from threshold-triggered crashes. The original trigger type remains for auditability, and the inbox can surface both the original harsh event type and crash tags on promoted events. Events that reviewers determine are not crashes continue as the original harsh event type.
  Crashes detected after a trip ends—while the Vehicle Gateway remains powered on—follow the same review path as on-trip crashes and surface in the **Safety Inbox** when confirmed. Off-trip crashes cannot be coached because driver assignment requires an active trip.
  Driving events that do not exceed 2 g and are not promoted through review do not automatically become crash events; however, if the vehicle was on a trip, you can request video retrieval in order to Manually Generate a Safety Event.
  Common scenarios that may surface as crash events include:
  - **Low impact events**: Low-speed impacts such as parking lot collisions or sideswipes may not exceed 2 g and therefore may not trigger accelerometer-only crash detection; some may still be captured when video-based review promotes a harsh brake, harsh acceleration, or harsh turn to a crash.
  - **False positive crashes**: High g-force events not caused by collisions—such as a kicked Vehicle Gateway, pothole, or reverse hookup—may initially surface as crashes.

  After a crash event surfaces, the system automatically evaluates the event for accuracy. If the event is later determined not to be a collision, it is moved to the **Dismissed** folder unless a user has already changed the event status.
- **High Speed Power Loss**: When a Vehicle Gateway loses power while the vehicle is traveling above your configured speed threshold, Samsara surfaces a **High Speed Power Loss** event in the **Safety Inbox**. This detection is a fallback for severe impacts that prevent the gateway from logging a standard crash event. High Speed Power Loss is off by default and is configured separately from crash sensitivity. To enable it, see Configure Harsh Event Settings.

> **Note**
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> High Speed Power Loss events may include limited or no video because the Vehicle Gateway lost power during the incident.

- **Harsh acceleration**: Triggers a 10-second video when the g-force threshold is exceeded. On supported CM3x and gateway combinations, harsh event detection applies stronger filtering for stationary or very low-speed harsh acceleration to reduce false positives. On CM11, CM12, and CM22 cameras, some very low-speed harsh accelerations may still be filtered and not generated.
- **Harsh brake**: Triggers a 10-second video when the g-force threshold is exceeded. A **Very Low** sensitivity level is recommended only for fleets with vehicles that produce frequent braking events, such as dumpster trucks or construction vehicles.
- **Harsh turn**: Triggers a 10-second video when the g-force threshold is exceeded.